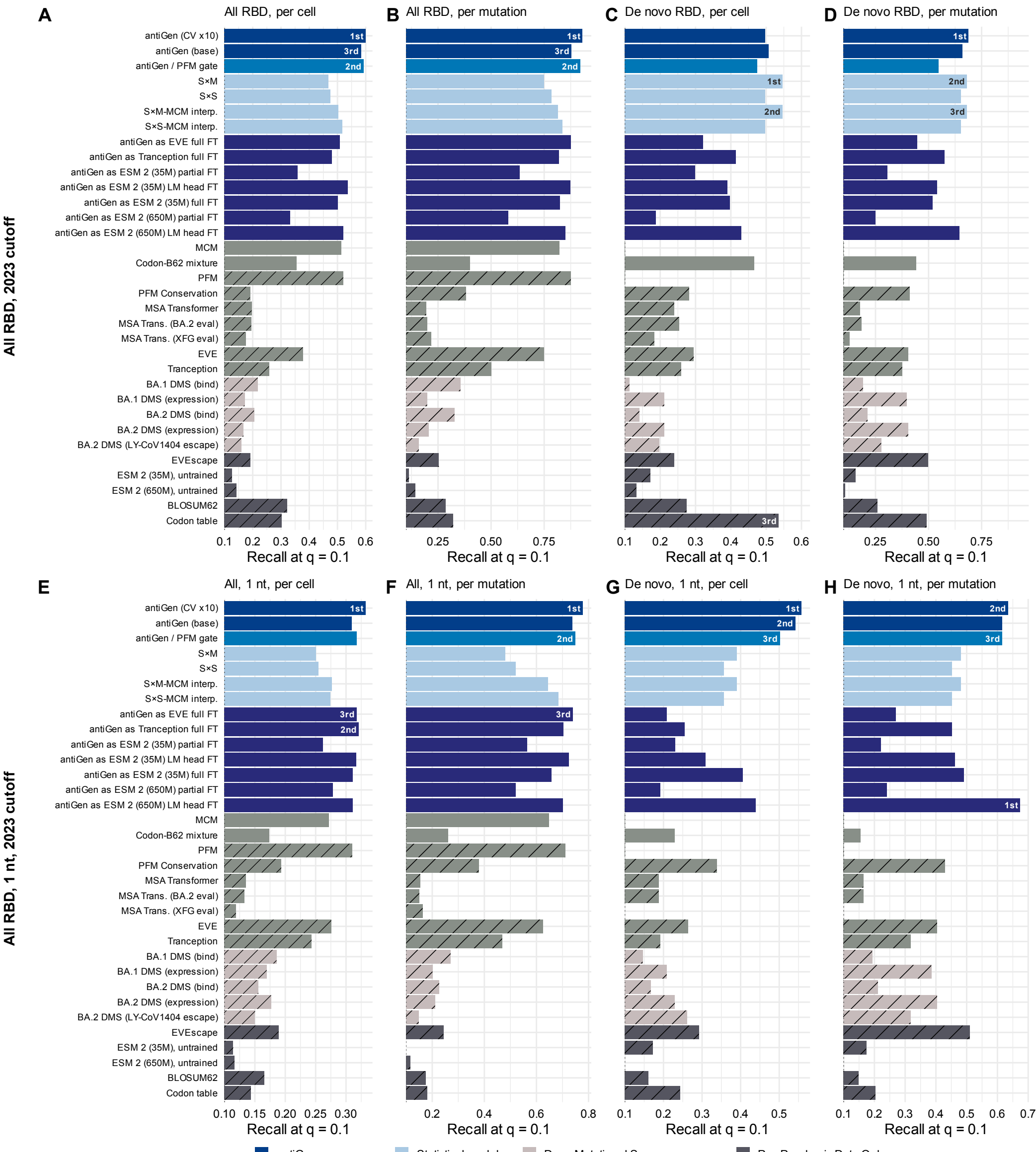

A
All RBD, per cell
B
All RBD, per mutation
C
De novo RBD, per cell
D
De novo RBD, per mutation
All RBD, 2023 cutoff
E
All, 1 nt, per cell
F
All, 1 nt, per mutation
G
De novo, 1 nt, per cell
H
De novo, 1 nt, per mutation
All RBD, 1 nt, 2023 cutoff
antiGen (CV x10)
antiGen (base)
antiGen / PFM gate
S×M
S×S
S×M-MCM interp.
S×S-MCM interp.
antiGen as EVE full FT
antiGen as Tranception full FT
antiGen as ESM 2 (35M) partial FT
antiGen as ESM 2 (35M) LM head FT
antiGen as ESM 2 (35M) full FT
antiGen as ESM 2 (650M) partial FT
antiGen as ESM 2 (650M) LM head FT
MCM
Codon-B62 mixture
PFM
PFM Conservation
MSA Transformer
MSA Trans. (BA.2 eval)
MSA Trans. (XFG eval)
EVE
Tranception
BA.1 DMS (bind)
BA.1 DMS (expression)
BA.2 DMS (bind)
BA.2 DMS (expression)
BA.2 DMS (LY-CoV1404 escape)
EVEscape
ESM 2 (35M), untrained
ESM 2 (650M), untrained
BLOSUM62
Codon table
Recall at q = 0.1
Model type
antiGen
antiGen gated hybrid
Statistical models
antiGen Finetunes
Deep Mutational Scans
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous